

Software Requirements Specification

for

ValeCare

Version 1.0

Prepared by WG4/SRHR Team

Formed to research SRHR for adolescent girls and young women, and built ValeCare as a result
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Table of Contents

Revision History.....	3
1. Introduction.....	3
1.1 Purpose.....	3
1.2 Document Conventions.....	3
1.3 Intended Audience and Reading Suggestions.....	3
1.4 Product Scope.....	3
1.5 References.....	3
2. Overall Description.....	3
2.1 Product Perspective.....	3
2.2 Product Functions.....	3
2.3 User Classes and Characteristics.....	4
2.4 Operating Environment.....	4
2.5 Design and Implementation Constraints.....	4
2.6 User Documentation.....	4
2.7 Assumptions and Dependencies.....	4
3. External Interface Requirements.....	4
3.1 User Interfaces.....	4
3.2 Hardware Interfaces.....	5
3.3 Software Interfaces.....	5
3.4 Communications Interfaces.....	5
4. System Features.....	5
4.1 Welcome and Onboarding.....	5
4.2 Account and Guest Access.....	5
4.3 Home Dashboard.....	6
4.4 Education and Learning.....	6
4.5 AI Health Assistant and Symptom Checker.....	7
4.6 Provider Search and Appointment Booking.....	7
4.7 Counselor Support and Emergency Assistance.....	7
4.8 Menstrual Cycle Tracker.....	8
4.9 Pregnancy, Fertility, and Family Planning.....	8
4.10 Notifications, Profile, Settings, and Account Management.....	8
5. Other Nonfunctional Requirements.....	9
5.1 Performance Requirements.....	9
5.2 Safety Requirements.....	9
5.3 Security Requirements.....	9
5.4 Software Quality Attributes.....	9
5.5 Business Rules.....	9
6. Other Requirements.....	9
Appendix A: Glossary.....	10
Appendix B: Analysis Models.....	10
Appendix C: To Be Determined List.....	10

Revision History

Name	Date	Reason For Changes	Version
Joy Chepkorir Bett	6 August 2026	Initial submission	1.0

1. Introduction

1.1 Purpose

This document specifies the software requirements for ValeCare, version 1.0, a web based platform that gives adolescent girls and young women private access to sexual and reproductive health information and services. It covers the complete system: the frontend, built in React and Vite as an installable Progressive Web App, and the backend, built in Django with Django REST Framework.

1.2 Document Conventions

Every functional requirement is written as a statement of what the system shall do and is uniquely identified with a requirement ID. Requirements in Section 4 are numbered REQ-x.y, where x is the system feature number and y is the requirement within that feature. Requirements in Section 5 are numbered NFR-z, in the order they appear. No priority scheme beyond the High, Medium, and Low ratings given for each system feature is used in this document.

1.3 Intended Audience and Reading Suggestions

This document is written for the development team building ValeCare and for course facilitators and reviewers assessing the project. Sections 1 and 2 give a general overview and are a good starting point for a first read. Section 4 is the most detailed and is the primary reference for a developer implementing a specific feature. Section 5 covers requirements that apply across the whole system rather than to one feature.

1.4 Product Scope

ValeCare lets a user browse verified sexual and reproductive health content, ask an AI assistant health questions and check symptoms, search for youth friendly healthcare providers and request an appointment, log her menstrual cycle and pregnancy, and reach a trusted safe contact, all without being required to create an account. Every feature works fully for an anonymous guest. An optional, lightweight account, a username and a passphrase, is the only way to keep a history of that activity across visits and devices. The system does not collect a user's name, phone number, or email address at any point.

1.5 References

- ValeCare Frontend Features Specification, React and Vite Implementation
- ValeCare Backend Features Document, Django REST Framework Implementation
- IEEE Std 830, Recommended Practice for Software Requirements Specifications

2. Overall Description

2.1 Product Perspective

ValeCare is a new, standalone product rather than an extension of an existing system. The frontend is a Progressive Web App that runs in a mobile or desktop browser and can be installed to a device's home screen, working offline for previously cached content. The backend is a set of Django REST Framework applications, one per feature area, that the frontend communicates with over a JSON API.

2.2 Product Functions

At a high level, ValeCare provides the following functions, described in full in Section 4:

- Welcome and onboarding with language selection
- Optional account creation and anonymous guest access

- A home dashboard summarizing recommended content, nearby providers, and reminders
- An education library of verified sexual and reproductive health content
- An AI health assistant and symptom checker
- Provider search and appointment booking
- A trusted safe contact connection and offline emergency guidance
- A menstrual cycle tracker
- Pregnancy, fertility, and family planning information and tracking
- Account, notification, and privacy settings

2.3 User Classes and Characteristics

Guest User: uses the application without creating an account. Has full access to every feature, and nothing about her visit is saved once she leaves.

Registered User: has created a username and passphrase. Has the same access as a guest, plus a saved history of chat sessions, bookmarks, and cycle or pregnancy logs that follows her to any device she signs in on.

Content Moderator: reviews and verifies educational content and emergency guidance before it is made available to users, through the Django administrative interface. Not a user of the public facing application.

2.4 Operating Environment

The frontend runs in modern mobile and desktop web browsers and is installable as a Progressive Web App. The backend runs on Django with Django REST Framework and a PostgreSQL database. The system is designed to remain usable on slow or intermittent mobile connections common in the regions it targets.

2.5 Design and Implementation Constraints

- No name, phone number, or email address may be collected or stored anywhere in the system.
- Every user facing screen must be available in a wide set of languages native to Africa, alongside English, French, and Portuguese.
- The interface must meet WCAG 2.1 AA accessibility guidelines.
- Content and layout must remain usable on low bandwidth connections and lower end devices.
- No feature may filter or restrict content based on the age of the user.

2.6 User Documentation

In-app guidance is limited to a short privacy explanation shown during onboarding and plain language empty states and error messages throughout the application. No separate user manual is planned for the initial release.

2.7 Assumptions and Dependencies

- Users have access to a smartphone or computer with a modern web browser and at least intermittent internet access.
- Provider distance filtering depends on the user's device supplying an approximate location.
- Educational content and emergency guidance are written and verified by the project's content team before publication.

3. External Interface Requirements

3.1 User Interfaces

The application is organized around a persistent header, a main content area, a floating emergency action button reachable from any screen, and navigation that adapts between a bottom bar on mobile and a side panel on larger screens. Screens are grouped by feature: onboarding, dashboard, education library, assistant chat, provider search, tracker, and settings. All screens follow a shared visual style and plain language tone.

3.2 Hardware Interfaces

The system has no dedicated hardware interface. It runs on any device capable of running a modern web browser, including lower specification Android devices, and does not depend on device sensors beyond optional location access for provider search.

3.3 Software Interfaces

- The frontend communicates with the backend over a JSON REST API served by Django REST Framework.
- The backend stores application data in a PostgreSQL database.
- The frontend uses Leaflet for map display and React i18next for translation.
- Offline caching is handled through IndexedDB and a service worker registered by vite-plugin-pwa.

3.4 Communications Interfaces

All communication between the frontend and backend takes place over HTTPS. The system does not send SMS messages, emails, or push notifications that reference sexual or reproductive health by name, in keeping with the project's privacy requirements.

4. System Features

This section lists the functional requirements for ValeCare, organized by the ten system features that make up the product. Each feature names the backend application responsible for it.

4.1 Welcome and Onboarding

Description and Priority

The first screen a user sees. It introduces the app, lets her pick a language, and offers three ways forward: continue as guest, sign in, or create an account. Priority: High. Backend: accounts application.

Stimulus/Response Sequences

- User opens the app for the first time. System displays the language options and the three entry choices.
- User selects a language. System stores the selection and displays the welcome screen in that language.
- User selects an entry path. System navigates to the corresponding guest, sign in, or account creation flow.

Functional Requirements

REQ-1.1 The system shall let a user select a display language before entering the application.

REQ-1.2 The system shall present three entry paths on the welcome screen: continue as guest, sign in, and create an account.

REQ-1.3 The system shall display a short privacy notice describing anonymous access before the user proceeds past onboarding.

4.2 Account and Guest Access

Description and Priority

Lets a user access every feature anonymously as a guest, or create a lightweight account so her history follows her across visits and devices. Priority: High. Backend: accounts application.

Stimulus/Response Sequences

- User selects Continue as Guest. System generates a device token and grants full access with no further input required.
- User selects Create Account and submits a username and passphrase. System creates the account and displays a one time recovery code.
- User selects Sign In and submits her username and passphrase. System verifies the credentials and links the device to the account.

- User selects Recover Access and submits her username and recovery code. System verifies them and lets her set a new passphrase.

Functional Requirements

- REQ-2.1** The system shall allow a user to start a guest session identified only by a device generated token, with no other information collected.
- REQ-2.2** The system shall allow a user to create a registered account using a username and a passphrase only.
- REQ-2.3** The system shall generate and display a one time recovery code immediately after a registered account is created.
- REQ-2.4** The system shall allow a registered user to sign in on any device using her username and passphrase.
- REQ-2.5** The system shall allow a user to reset a forgotten passphrase using her username and her recovery code, and shall issue a new recovery code after a successful reset.
- REQ-2.6** The system shall allow a registered user to change her language, username, or passphrase at any time.

4.3 Home Dashboard

Description and Priority

The landing screen after entry, summarizing recommended content, nearby providers, and open reminders. Priority: Medium. Backend: education, providers, followup, and emergency applications.

Stimulus/Response Sequences

- User completes onboarding or signs in. System loads and displays recommended content, nearby providers, and pending reminders.

Functional Requirements

- REQ-3.1** The system shall display recommended educational content on the dashboard.
- REQ-3.2** The system shall display nearby verified providers on the dashboard based on the user's approximate location.
- REQ-3.3** The system shall display the current user's pending follow up reminders on the dashboard.

4.4 Education and Learning

Description and Priority

A learning hub for browsing and reading verified sexual and reproductive health content, open to every user regardless of age. Priority: High. Backend: education application.

Stimulus/Response Sequences

- User opens the education library and selects a category. System displays the matching content items.
- User opens a content item. System displays the full article and, for a registered user, shows bookmark and progress controls.
- Registered user bookmarks an item. System saves the bookmark to her account.

Functional Requirements

- REQ-4.1** The system shall allow any user, guest or registered, to browse and read verified sexual and reproductive health content filtered by category and language.
- REQ-4.2** The system shall not filter or restrict content based on the age of the user.
- REQ-4.3** The system shall allow a registered user to bookmark a content item for later reading.
- REQ-4.4** The system shall allow a registered user to track her reading progress on a content item.
- REQ-4.5** The system shall not save bookmarks or reading progress for a guest user beyond the current session.

4.5 AI Health Assistant and Symptom Checker

Description and Priority

A conversational assistant for sexual and reproductive health questions, paired with a symptom checker. Priority: High. Backend: triage application.

Stimulus/Response Sequences

- User opens the assistant and asks a question. System returns a reply and indicates whether it is drawn from verified content.
- User starts the symptom checker and answers a series of prompts. System returns guidance appropriate to the symptoms described.

Functional Requirements

REQ-5.1 The system shall provide a conversational assistant where a user can ask sexual and reproductive health questions.

REQ-5.2 The system shall provide a symptom checker that records a user's plain language description of her symptoms.

REQ-5.3 The system shall visibly label an assistant reply that is sourced from verified content.

REQ-5.4 The system shall save a registered user's chat history for later reference on the dashboard.

REQ-5.5 The system shall not retain a guest user's chat history beyond the current session.

4.6 Provider Search and Appointment Booking

Description and Priority

A directory of verified, youth friendly providers with a simple booking flow. Priority: High. Backend: providers and appointments applications.

Stimulus/Response Sequences

- User opens provider search and applies filters. System returns a filtered list of providers.
- User selects a provider and requests an appointment. System creates the appointment and returns a reference code.

Functional Requirements

REQ-6.1 The system shall allow a user to search for healthcare providers filtered by gender, distance, and faith sensitivity.

REQ-6.2 The system shall allow a user to submit a rating and review for a provider she has visited.

REQ-6.3 The system shall allow a user to request an appointment with a provider.

REQ-6.4 The system shall identify a booked appointment to the provider using a reference code rather than the user's identity.

4.7 Counselor Support and Emergency Assistance

Description and Priority

Connects a user to one trusted safe contact and provides offline capable emergency guidance. Priority: High. Backend: support and emergency applications.

Stimulus/Response Sequences

- User opens her safe contact chat and sends a message. System delivers the message and displays the response when it arrives.
- User taps the emergency button from any screen. System immediately displays cached emergency guidance, with or without a connection.

Functional Requirements

REQ-7.1 The system shall let a user connect with one verified trusted safe contact through a dedicated chat thread.

REQ-7.2 The system shall provide a persistent, one tap emergency guidance button reachable from every screen.

REQ-7.3 The system shall cache emergency guidance content locally so it remains available without an internet connection.

REQ-7.4 The system shall refresh cached emergency guidance only when a newer version is published on the server.

4.8 Menstrual Cycle Tracker

Description and Priority

Lets a user log her menstrual cycle and see predictions for her next cycle and fertile window. Priority: Medium. Backend: tracker application.

Stimulus/Response Sequences

- User logs a new cycle entry. System saves the entry and updates her prediction for the next cycle.

Functional Requirements

REQ-8.1 The system shall allow a user to log the start date, end date, flow level, symptoms, and mood for a menstrual cycle.

REQ-8.2 The system shall estimate a user's next period and fertile window from her logged cycle history.

REQ-8.3 The system shall save a registered user's cycle history on the server so it is available on any device she signs in on.

REQ-8.4 The system shall keep a guest user's cycle entries on the local device only, without sending them to the server.

4.9 Pregnancy, Fertility, and Family Planning

Description and Priority

Combines educational content on pregnancy and family planning with a personal pregnancy journal. Priority: Medium. Backend: education and tracker applications.

Stimulus/Response Sequences

- User opens the pregnancy section and adds a journal entry. System saves the entry, for a registered user, and displays the computed current week.

Functional Requirements

REQ-9.1 The system shall provide educational content on pregnancy, fertility, and contraceptive options to any user.

REQ-9.2 The system shall allow a registered user to maintain a pregnancy journal with a computed current week of pregnancy.

REQ-9.3 The system shall keep a guest user's pregnancy journal entries on the local device only, without sending them to the server.

4.10 Notifications, Profile, Settings, and Account Management

Description and Priority

Lets a user manage her preferences and, if registered, her account. Priority: Medium. Backend: accounts application.

Stimulus/Response Sequences

- User opens settings and changes her language. System applies the change immediately.
- Registered user requests account deletion. System asks her to re-enter her passphrase, shows a final warning, and deletes the account only after she confirms.

Functional Requirements

REQ-10.1 The system shall allow any user to change her selected language and clear locally stored data at any time.

REQ-10.2 The system shall allow a registered user to manage her notification preferences.

REQ-10.3 The system shall allow a registered user to request deletion of her account.

REQ-10.4 The system shall require the user to re-enter her passphrase and confirm a final warning before an account deletion is carried out.

REQ-10.5 The system shall permanently remove a deleted account's saved history, including chat sessions, bookmarks, and tracker entries.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

NFR-1 The system shall load its initial screen within an acceptable time on a low bandwidth mobile connection, through lazy loading and compressed assets.

NFR-2 The system shall keep previously cached education and emergency content available for reading without an active internet connection.

5.2 Safety Requirements

NFR-3 The system shall direct a user toward urgent care guidance whenever the symptom checker or AI assistant identifies a possible medical emergency.

NFR-4 The system shall keep the emergency guidance button reachable from every screen so a user is never more than one tap away from help.

5.3 Security Requirements

NFR-5 The system shall not collect a user's name, phone number, or email address at any point.

NFR-6 The system shall store passphrases and recovery codes in hashed form only, never as plain text.

5.4 Software Quality Attributes

NFR-7 The system shall conform to WCAG 2.1 AA accessibility guidelines, including keyboard navigation and screen reader support.

NFR-8 The system shall present information in plain language suitable for a teenage reader, avoiding clinical terminology.

NFR-9 The system shall support English, French, Portuguese, and a broad set of languages native to Africa, including Swahili, Igbo, Yoruba, Hausa, Amharic, Zulu, Xhosa, Shona, Somali, Twi, Wolof, Lingala, Kinyarwanda, and Afrikaans.

5.5 Business Rules

- Only a registered user may access saved history features, including bookmarks, chat history, and tracker entries.
- Only a verified provider may appear in provider search results.
- Only content marked verified may be labeled as a verified source by the AI assistant.

6. Other Requirements

Internationalization: the interface must render correctly in every supported language, with translated strings maintained centrally rather than hardcoded per screen.

Data handling: since the system collects no name, phone number, or email address, standard data protection obligations are reduced in scope, but stored content, such as chat history and tracker entries, must still be handled as sensitive personal health data and deleted completely on account deletion.

Database: the system uses PostgreSQL and is a new deployment with no legacy data migration requirements.

Appendix A: Glossary

- SRHR: sexual and reproductive health rights.
- AGYW: adolescent girls and young women.
- PWA: Progressive Web App, a website that can be installed to a device and work offline.
- Guest session: an anonymous visit identified only by a device generated token, with nothing saved once it ends.
- Registered account: an optional account identified by a username and passphrase, used to save history across visits.
- Recovery code: a one time code shown at account creation, used to reset a forgotten passphrase.

Appendix B: Analysis Models

Detailed data models for every backend application, including field level definitions for each entity, are documented in the ValeCare Backend Features Document. Screen level flows and component breakdowns for every feature are documented in the ValeCare Frontend Features Specification. Both should be read alongside this specification for implementation detail beyond the requirement statements above.

Appendix C: To Be Determined List

- TBD-1: Final team member names and submission date for the title page and revision history.
- TBD-2: Specific performance benchmarks, such as target load times, once testing on representative devices is available.
- TBD-3: Formal data flow or entity relationship diagrams, if required by the course facilitator, to supplement Appendix B.